

Repressive Democracies: the Relationship Between State's Repression Severity, Protests, and Policy Change.

Ignacio Urbina*

Ph.D. Candidate in Political Science, Stony Brook University

September 11, 2026

1 Introduction.

Scholarly work that inquiries into state repression's consequences on mobilization dynamics have revealed puzzling results (Davenport 2007a). On the one hand, some studies show that threats of state repression may deter further mobilization by increasing its cost to protesters (Chenoweth, Perkosi, and Kang 2017; della Porta 1995; Christopher M Sullivan 2016). In opposition, others reveal that repression backfires, meaning that protesters who are victims of violence become radicalized (Adam-Troian, Çelebi, and Mahfud 2020; Christopher Michael Sullivan and Davenport 2017). Moreover, when one thinks of repression severity as a policy variable that the states choose, it becomes clear that repression and dissent are strategically interlinked.

The study of this phenomenon, framed as the “protest-repression” nexus (Davenport 2007a; Curtice and Behlendorf 2021), has been developed primarily in the context of autocracies. An empirical strand does examine protest in democracies (Carey 2006; Nam 2007; Machado, Scartascini, and Tommasi 2011; Su 2015; Aytaç, Schiumerini, and Stokes 2017), but it is largely institutional, asking what makes protest more or less likely across democratic settings rather than how a state chooses the severity with which it meets protest. This is especially true when theoretical or micro-founded approaches have been put forth to explain empirical patterns in cross-country

*Email: ignacio.urbina@stonybrook.edu. Website: www.ignacio-urbina.com.

perspectives (Bueno de Mesquita 2010; Bueno de Mesquita and Shadmehr 2023; Gehlbach, Sonin, and Svolik 2016; Ginkel and Smith 1999; Little 2016). Yet, there are several reasons to think that this phenomenon deserves a closer and deeper study in democratic regimes. First, democracies are naturally well-suited environments for individuals that consider protesting if they hold grievances against political elites or other societal groups, because electoral competition incentivizes political elites to attend to such grievances.

Second, in democracies, for incumbent political elites, mass protests are costly for two reasons. On the one hand, a massive protest may force politicians to change policies they might not want to alter (Bosi, Giugni, and Uba 2016; Klein and Regan 2018). For example, gridlocked constitutional dispositions that benefit elites, such as the tax code, the electoral system, or others. Additionally, massive protests imply that the state needs to spend resources to “oversight” them. Most of the time, this implies having a specialized anti-protest or anti-riot police corps, which possibly implies expensive budgets to sustain them.

In democracies, the severity of repression is a complex policy that is implicitly written in the institutional and legal framework. For democratic regimes, leaders do not command repressive bureaus to repress citizens in particular ways. Nevertheless, defining a given legal structure that allows repressive operatives to exert higher or lower degrees of physical or physiological damage to dissenter citizens implicitly sets the severity of the repression policy. That structure is slow to change; the model therefore treats the severity policy as chosen before mobilization and held fixed while a protest runs, an assumption I state and qualify in the next section. Repressive bureaus likely know the rules of engagement they can follow, imprinted on the formal institutional framework, which allows them to use different levels of violence on citizens. Since savvy politicians know that protests are always possible in the foreseeable future, they will think very hard when setting the repression severity policy, weighing costs and benefits.

In this paper, I develop a game-theoretic model that integrates the state’s decision regarding the severity of the repression policy and the citizens’ decision to protest motivated by a prospect of policy change. The model has a number of essential features. First, citizens face a collective action problem when they decide to protest: the protest changes the policy only if participation reaches a critical threshold. Second, that critical threshold is not common knowledge. Citizens hold noisy private assessments of how large the protest must be to force the policy change, and

they must decide whether to participate under this strategic uncertainty. This is the global games structure: as in the canonical regime change game, the incomplete information about the fundamental delivers a unique equilibrium in threshold strategies, where the complete-information version of the same game exhibits multiple equilibria. Third, given that the state would prefer to keep the status quo, citizens' and state incentives are directly opposed. Thus, the state decides on a repression severity policy that imposes costs on individuals who choose to protest. Importantly, although a higher severity of repression may deter protests, setting a repression severity policy is costly for the state.

The rest of the paper is organized as follows. First, I offer a brief literature review that informs the assumptions and context of the paper, including the global games literature on protest and regime change that supplies the machinery for the citizens' stage. Next, I develop the citizens' participation decision as a global game of policy change, presenting the complete-information benchmark first to make the role of the incomplete information explicit. Then, I integrate the state's decision on the repression severity policy into a two-stage sequential game. Furthermore, I discuss the implications of the model, underlining insights for empirical research. Finally, I present the conclusions of the paper.

2 Literature review.

2.1 State repression, political attitudes, and social movements.

The study of state repression has a long and vast literature. Broadly, the act of subduing or coercing individuals or groups that hold claims to or challenge political actors or the status quo, using institutional or physical force, most commonly using violent means, is known as political or "state" repression (deMeritt 2016). Many empirical regularities have been documented and different theoretical approaches have been utilized to understand why states repress citizens, and how this influences their political behavior, especially their participation in protests. It should not be surprising that this body of research is overwhelmingly focused on authoritarian regimes or flawed democracies. The intrinsic characteristics of such regimes, such as the lack of accountability of political leaders, and the absence of freedom of speech or competitive elections, enable rulers to deter opposition using violent means.

Thus, comparative research persistently revealed that democratic countries exhibit lower levels of state repression (Davenport 2007a, 2007b; Davenport and Armstrong 2004). Yet, state repression, especially in the form of violent coercion perpetrated by state agents, or “repressive operatives,” is not uncommon in developed democracies. In fact, state violence, especially police brutality, has become increasingly ubiquitous across some democracies (Belur 2010; Whitehead, Fair, and Payne 2010; Worden 2015). These expressions of violence tend to be very salient and capture the attention of the public. For example, there is evidence that after the killing of George Floyd at the hands of the police, support for the police substantially fell (Reny and Newman 2021). Moreover, victims of police repression may experience a range of emotional states such as anxiety, depression, anger, and fear (Bor et al. 2018; Çelebi, Adam-Troian, and Mahfud 2020; Cooper et al. 2004; J. E. DeVlyder et al. 2017; J. DeVlyder, Fedina, and Link 2020).

Yet not all citizens perceive the use of force by repressive bureaus in the same fashion. For example, in the U.S., studies have persistently revealed that conservatives have a propensity to support the police at higher rates than liberals (Ekins, 2016; Fingerhut, 2017; Hindelang, 1974; Zamble & Annesley, 1987). To the extent that voters have directional motivations and are affective partisans, they may perceive, process, and evaluate state repression in different ways. For example, attitudes about the police are racially divided (Ekins, 2016; Fingerhut, 2017).

One essential question in the literature is what the consequences of state repression in the context of social movements are. Studies have generally referred to this problem as the “punishment puzzle” (Davenport 2007a). On the one hand, studies have shown that repression discourages mass and peaceful protests but stirs radicalization (della Porta 1995). In contrast, there is also evidence that suggests that repression incites conflict escalation and radicalization, thus provoking higher levels of dissent and mobilization (Adam-Troian, Çelebi, and Mahfud 2020; Christopher Michael Sullivan and Davenport 2017).

2.2 Empirical regularities and theoretical foundations.

The literature has studied several determinants of state repression, revealing that regime type, economic development, population size, intrastate civil conflicts, and degrees of grievances and dissent systematically explain state repression (Davenport 2007b; Nordås and Davenport 2013; Christopher M Sullivan 2016; Christopher Michael Sullivan and Davenport 2017). For example,

Nordås and Davenport (2013) argue that the demographic structure of the population is a key determinant of the level of state repression. To the extent that countries exhibit larger cohorts of “youth bulges”, people between the ages 15–24, relative to the overall adult population, then states respond by increasing repression. The rationale for this response is that if states anticipate a higher level of dissent, then they preemptively increase repression to induce deterrence.

Moreover, Chenoweth, Perkoski, and Kang (2017) summarize several consistent findings in the literature. To begin with, a systematic finding is that dissent antecedes state repression in most settings. This finding has been labeled as the “law of coercive responsiveness” (Davenport 2007a). Also, democratic countries tend to exhibit lower levels of repression. Note that the authors also discuss the “punishment puzzle” mentioned above. Overall, they stress that repression and dissent, in the form of protests or other tactics, are two complex interlinked processes.

On the side of formal models, several approaches have been attempted to shed light on the theoretical underpinnings of the relationship between state repression, protests, and regime or policy change. Although there is not a definite consensus for what is the best modeling approach. Ritter (2014) notes that models have chosen different main objects of investigation that, if taken separately, miss some of the key strategic features of the relation between repression and dissent. For example, some studies model the determinants of regime change or dissent in general as a process of an ongoing or expected conflict. One way to approach this is that citizens to protest to alter the status quo, which depends on the state’s concessions. For example, the level of violence of a revolutionary vanguard may signal the level of dissent in society, such that citizens’ cooperation is provoked (Bueno de Mesquita 2010). Other formal models have stressed the role of uncertainty and asymmetries of information in government types (Pierskalla 2010). For example, a “strong” state may be willing to costly repress citizens to signal its resolve. Over repeated play, citizens protests and learn about the type of government they face. Closest to the present paper in timing, De Jaegher and Hoyer (2019) model *preemptive* repression: a government sets repressive capacity before protest occurs, and dissidents respond by investing in countermeasures. Their model delivers both sides of the punishment puzzle – repression deters when the benefit of protest is large relative to the cost of countermeasures, and backfires when it is small – but it does so in a complete-information coordination game between two pivotal dissidents, with the resulting multiplicity resolved by the relative size of the equilibria’s basins of attraction.

Thus, as Ritter (2014) notes, some models study the conditions that ignite dissent, and others study how the state chooses the severity of repression. But, if these studies take these two processes as separate entities, then an important part of the puzzle is missed. Hence, she proposes a model that jointly considers the conditions that spark dissent, or protest, and how the state chooses the severity of repression. The strategic setting is modeled by a bargaining model of policy provision, and its effect on political incumbent survival. The results show that an increase in the likelihood of survival of the political leaders decreases the level of dissent and repression. Yet, an increase in the likelihood of survival of the political leaders increases the severity of repression and the severity of dissent.

2.3 Global games of protest and regime change.

The machinery this paper borrows comes from the global games literature on coordination under incomplete information (Carlsson and van Damme 1993; Morris and Shin 1998, 2003). In the canonical global game of regime change, a continuum of citizens simultaneously decide whether to attack a regime that collapses if and only if the mass of attackers exceeds the regime's strength, which is an uncertain fundamental about which citizens receive noisy private signals. Under complete information about the fundamental, the game exhibits multiple equilibria in an intermediate range of regime strength; the incomplete information selects a unique equilibrium in threshold strategies.

Two recent applications inform the present model directly. Bueno de Mesquita and Shadmehr (2023) develop “a standard global game of regime change” in which a continuum of citizens receive noisy private signals $x_i = \theta + \sigma \varepsilon_i$ about the regime's strength θ , and the rebellion succeeds if and only if the fraction of rebels exceeds θ . Their complete-information benchmark (their Proposition 1) shows that for intermediate strength “both equilibria coexist,” and they note that this multiplicity “makes empirical prediction difficult” and renders outcomes “insensitive to parameters of the model like the costs of rebellion”; their incomplete-information analysis (their Proposition 2) delivers a unique equilibrium in which the rebellion succeeds if and only if the fundamental is below a threshold, equal to $1 - c$ in their psychological rewards setting, where c is the cost of participation. They adopt “the standard conception of state repression as any action by the state that ‘raises the contenders’ cost of collective action’ (Tilly 1978, 100),” representing more

repression as a higher c , and they add a first stage in which the government chooses its repression level by minimizing the sum of the regime change threshold and a convex cost of repressive capacity. A key ingredient of their psychological rewards setting, which the present model adopts, is that the participation reward is contingent on the movement's success: building on Wood's (2003) notion of pleasure-in-agency, participants enjoy the psychological reward of having taken part only if the rebellion succeeds, and this success-contingency is what generates the strategic complementarity that the global game resolves.

Little (2016) distinguishes two coordination problems that potential protesters face: a "political coordination" problem, deciding whether to participate without knowing how many others are dissatisfied enough to join, and a "tactical coordination" problem, deciding when, where, and how to protest. His political coordination model also features a continuum of citizens with heterogeneous assessments and equilibria in cutoff strategies, and he shows that better tactical information "almost always lowers the effective cost of protest, leading to larger (and more cohesive) antiregime action." This distinction matters for the present model because the critical threshold citizens face here is precisely the kind of object about which communication technology conveys information, a point I return to in the empirical implications.

3 Modeling protests, repression, and policy change.

In this section, I develop a model of citizens who decide to protest to change a disliked policy. The citizens face a coordination and collective action problem: the protest successfully changes the policy only if citizens' participation reaches a given threshold. The threshold, moreover, is not common knowledge: it is an uncertain fundamental about which citizens hold noisy private assessments. On the other side, the state would prefer not to change the policy. To influence protest participation, it can set a repression severity policy that inflicts costs on individuals who decide to protest. Nevertheless, producing a given level of repression (i.e., a given number of victims of repression) is costly in resource utilization, and setting a severity of repression policy is costly also because of public opinion backlash consequences. In what follows, I begin by setting up the citizens' participation problem and its complete-information benchmark. Next, I solve the global game. Then, I propose the two-stage sequential game in which the state decides the severity

of the repression policy first, and the strategic implications of the model are inspected.

3.1 The citizens' participation problem.

There is a continuum of citizens of mass one, indexed by $i \in [0, 1]$. Each citizen chooses between participating in the protest (P) or not (NP); let $n_i \in \{0, 1\}$ indicate participation and let $N^* \in [0, 1]$ denote the realized mass of participants. The objective of protesters is to change a policy that has such a degree of gridlock that it is tough to change over traditional forms of representation. The protest is successful if participation reaches a critical threshold τ : the policy changes if and only if $N^* \geq \tau$.

The payoff primitives are as follows. A change in the policy provides to all individuals in the society a collective and indivisible benefit of B_p (i.e., the policy change provides public good benefits). Individuals who participate may get an expressive or emotional benefit d_p ; for example, they may feel that fighting for a just cause is good in and of itself. Participants also bear the expected cost of repression $\gamma_r \cdot c_r$, where c_r is the severity of repression (the disutility inflicted on a repressed protester) and $\gamma_r \in [0, 1]$ is the probability that a participant is a victim of state repression. Both c_r and γ_r are policy variables for the state.

Two features of the continuum setting must be made explicit, because they discipline how the primitives enter the participation margin.

First, no individual citizen is pivotal. In a finite-player version of this game, the term Δp_i , the change in the probability of reaching the threshold induced by one's own participation, would carry the public good benefit B_p into the individual calculus. In the continuum, a single citizen has measure zero, so $\Delta p_i = 0$: the citizen receives B_p when the protest succeeds whether or not she participated, and B_p cancels from the comparison between participating and abstaining. The free-riding logic is total here. This is not a defect of the machinery but a substantive statement it forces us to make: in a mass protest, the policy prize itself cannot be what moves the marginal participant, because her participation does not detectably move the prize.

Second, for participation to respond to the prospects of success at all, the expressive benefit must be success-contingent. I follow the psychological rewards setting of Bueno de Mesquita and Shadmehr (2023), which builds on Wood's (2003) pleasure-in-agency: the expressive reward d_p is enjoyed by a participant only if the movement succeeds. The satisfaction of having been part of

forcing the change is realized when the change is forced; a participant in a failed protest bears the costs and receives no expressive payoff. This success-contingency is a substantive assumption, and it is what generates the coordination problem: were d_p unconditional, participation would reduce to the comparison of d_p against $\gamma_r c_r$, independent of any belief about others, and the strategic structure would disappear. An unconditional expressive reward is a defensible alternative reading; the success-contingent reading is the one under which mass participation responds to the prospects of success, which is the phenomenon this paper models.

With these two features, the net payoff of participating relative to abstaining, for citizen i , is

$$u_i(P) - u_i(NP) = \mathbb{1}\{N^* \geq \tau\} \cdot d_p - \gamma_r \cdot c_r,$$

since $B_p \cdot \mathbb{1}\{N^* \geq \tau\}$ enters both sides and cancels. A citizen therefore participates when her belief that the protest will succeed, times the expressive reward, exceeds the expected repression cost.

3.2 The complete-information benchmark.

Suppose first, as a benchmark, that τ is common knowledge. Assume $0 < \tau \leq 1$, the interesting range, and $\gamma_r c_r < d_p$, so that participation is worthwhile when success is assured. Then two pure strategy equilibria coexist. If every citizen participates, $N^* = 1 \geq \tau$, the protest succeeds, and each participant receives $d_p - \gamma_r c_r > 0$; no citizen gains by deviating, since abstaining forfeits $d_p - \gamma_r c_r$ while leaving success (and hence B_p) unchanged. If no citizen participates, $N^* = 0 < \tau$, the protest fails, and a lone deviator would pay $\gamma_r c_r$ for a failed protest; so universal abstention is also an equilibrium. If instead $\gamma_r c_r > d_p$, universal abstention is the unique equilibrium: participation is dominated even under assured success.

This benchmark mirrors the complete-information benchmark of Bueno de Mesquita and Shadmehr (2023, their Proposition 1), and it inherits the two problems they identify: the multiplicity makes prediction difficult, and the equilibrium set is insensitive to the parameters, including the repression policy itself. One could respond by selecting an equilibrium, for instance the mixed strategy equilibrium, and reading its participation probabilities as frequencies. I instead take seriously that the critical threshold is a variable citizens cannot observe. Unlike in elections under plurality rule, where the threshold to win is common knowledge, the size a protest must

reach to force a policy change is written nowhere, and how citizens form beliefs about τ is central to any behavioral prediction. That observation becomes the engine of the model.

3.3 The global game.

Let the critical threshold τ be drawn by nature and unobserved by citizens. Citizens share an improper uniform prior over $\mathbb{R}$, the standard device in this literature to isolate strategic uncertainty (Morris and Shin 2003; Bueno de Mesquita and Shadmehr 2023).¹ Each citizen receives a noisy private signal

$$x_i = \tau + \sigma \varepsilon_i,$$

where the ε_i are i.i.d. with a log-concave density f and c.d.f. F with full support on $\mathbb{R}$, independent of τ , and $\sigma > 0$ scales the noise. A citizen with a low signal believes the required crowd is small and success likely; a citizen with a high signal believes success is out of reach.

A strategy is a mapping from signals to participation. Since the participation payoff is decreasing in the (believed) threshold, it is natural to look for equilibria in cutoff strategies: participate if and only if $x_i < \hat{x}$. Given such a cutoff, the mass of participants at true threshold τ is

$$N^*(\tau) = \Pr(x_i < \hat{x} \mid \tau) = F\left(\frac{\hat{x} - \tau}{\sigma}\right),$$

which is continuous and strictly decreasing in τ . The protest therefore succeeds if and only if τ is below a unique critical value $\hat{\tau}$, and the equilibrium is characterized by the same two conditions as in the canonical regime change game (Bueno de Mesquita and Shadmehr 2023, their equation (1)):

$$\text{(belief consistency / critical mass):} \quad F\left(\frac{\hat{x} - \hat{\tau}}{\sigma}\right) = \hat{\tau},$$

$$\text{(indifference of the marginal citizen):} \quad \Pr(\tau \leq \hat{\tau} \mid x_i = \hat{x}) \cdot d_p = \gamma_r c_r.$$

The first condition says that at the critical threshold the crowd is exactly critical. The second says that the citizen holding exactly the cutoff signal is indifferent between participating and

¹The improper prior delivers exact closed forms. The same characterization obtains for any smooth proper prior in the limit as the signal noise becomes small.

abstaining. Under the improper uniform prior, the posterior of τ given $x_i = \hat{x}$ is $\hat{x} - \sigma\epsilon$, so

$$\Pr(\tau \leq \hat{\tau} \mid x_i = \hat{x}) = 1 - F\left(\frac{\hat{x} - \hat{\tau}}{\sigma}\right) = 1 - \hat{\tau},$$

where the last equality substitutes the belief consistency condition. The indifference condition then collapses to $(1 - \hat{\tau})d_p = \gamma_r c_r$, and the equilibrium success threshold is

$$\hat{\tau}(c_r) = 1 - \frac{\gamma_r c_r}{d_p}, \quad \text{for } \frac{\gamma_r c_r}{d_p} \in [0, 1]. \quad (1)$$

This is the exact analog of the threshold $\theta^p = 1 - c$ in Bueno de Mesquita and Shadmehr (2023, their equation (2) and Proposition 2), with the normalized participation cost $\gamma_r c_r / d_p$ in the role of c . Proposition 1 collects the characterization.

Proposition 1 (Equilibrium of the citizens' stage). *Fix a severity policy with $\gamma_r c_r / d_p \in (0, 1)$. The citizens' stage has a unique equilibrium in monotone strategies: citizen i participates if and only if $x_i < \hat{x} = \hat{\tau} + \sigma F^{-1}(\hat{\tau})$, and the protest succeeds if and only if $\tau \leq \hat{\tau} = 1 - \gamma_r c_r / d_p$. If $\gamma_r c_r \geq d_p$, no citizen participates and the protest fails against every $\tau > 0$; if $c_r = 0$, every citizen participates and $\hat{\tau} = 1$. Standard arguments extend uniqueness beyond the monotone class.²*

The full proof is in the appendix. Three remarks.

First, the multiplicity is gone. For every value of the repression policy there is one prediction: the protest succeeds if and only if the realized threshold satisfies $\tau \leq \hat{\tau}(c_r)$. The comparative statics of the model are therefore unconditional statements, not statements contingent on which equilibrium is played or on a selected mixed strategy.

Second, the repression policy now moves the outcome smoothly. The deterrence derivative is

$$\frac{\partial \hat{\tau}}{\partial c_r} = -\frac{\gamma_r}{d_p}.$$

Each unit of severity shrinks the set of fundamentals at which the protest succeeds at a constant rate, inversely proportional to the expressive reward. Repression against a strongly motivated citizenry (high d_p) is blunt; against a weakly motivated one it is sharp. This is the global games

²See Morris and Shin (2003) and the technical discussion in Bueno de Mesquita and Shadmehr (2023, their footnote 8).

rendering of a familiar intuition: intrinsic motivation erodes the effectiveness of severity.

Third, the deterrence margin is governed by d_p alone, not by $d_p + B_p$. In a small-numbers setting where participation is pivotal (as in De Jaegher and Hoyer 2019, where the protest occurs only if both dissidents participate), full deterrence would require severity to climb over the policy prize itself, since the pivotal citizen weighs B_p directly. Here, full deterrence ($\hat{\tau} = 0$) is reached already at $c_r = d_p/\gamma_r$: severity never needs to climb over B_p , because the free-riding of the continuum has already removed B_p from the participation margin. This is a substantive, testable implication of the mass-mobilization margin, discussed further below.

For the state's problem it is convenient to work in the small-noise limit $\sigma \rightarrow 0$. In that limit the cutoff $\hat{x}$ converges to $\hat{\tau}$ and participation becomes sharp: $N^*(\tau) \rightarrow 1$ for $\tau < \hat{\tau}$ and $N^*(\tau) \rightarrow 0$ for $\tau > \hat{\tau}$. Protests that succeed are massive; protests that fail do not materialize. This stark prediction is an artifact of vanishing noise, and with $\sigma > 0$ participation falls continuously in τ ; the limit is used only to keep the state's objective in closed form.

3.4 The state's problem.

The state moves first and sets the repression severity policy, anticipating the unique equilibrium of the citizens' stage.

Assumption 1 (Severity is fixed within a protest episode). The state chooses c_r before mobilization begins and does not revise it while the protest unfolds. Severity is chosen freely between episodes; it is sticky within one.

The institutional reading above is what supports this: rules of engagement are written into legal structure and standing procedure, and changing them takes longer than a protest lasts. The assumption is also what makes the game sequential; without it the state and the citizens move simultaneously, and severity loses its commitment value.

De Jaegher and Hoyer (2019) take the other branch: their government observes how far dissidents have invested in countermeasures and adapts its repression level accordingly, and the delay before it can do so is a parameter of their analysis. This model is the limit of that setup in which the delay exceeds the length of a protest episode. Which limit applies is an empirical question about the institution. Severity written into legal and doctrinal frameworks moves slowly;

repressive capacity that can be surged within days does not, and there the fixed-policy assumption would be the wrong one. I fix $\gamma_r = 1$ to focus on c_r as the policy variable, and I assume the following cost function for the production of repression:

$$f(c_r, N^*) = \theta_0 \cdot N^* + \theta_1 c_r^2,$$

where θ_0 is the parameter that influences the marginal cost of repressing one additional protester (e.g., costs in repressive inputs or equipment) and θ_1 is the marginal cost of increasing the severity of repression (e.g., public opinion marginal costs). The state keeps B_s if the protest fails and loses it if the protest succeeds.

The state evaluates its policy ex ante, before nature draws τ . For this evaluation the improper prior is not usable, and I endow the state with a proper planning prior $\tau \sim U[0, 1]$, following the same logic by which Bueno de Mesquita and Shadmehr (2023) give their government a proper distribution over the fundamental at its investment stage.³ Under this prior and the small-noise limit, the probability that the protest succeeds is $\Pr(\tau \leq \hat{\tau}) = \hat{\tau}(c_r)$, and the expected mass of protesters is $E[N^*] = \hat{\tau}(c_r) \cdot 1 = \hat{\tau}(c_r)$, since in the limit the protest is total exactly when it succeeds. The state's expected utility is

$$u_S = (1 - \hat{\tau}(c_r))B_s - \theta_0 \hat{\tau}(c_r) - \theta_1 c_r^2 = \frac{c_r}{d_p} B_s - \theta_0 \left(1 - \frac{c_r}{d_p}\right) - \theta_1 c_r^2,$$

substituting equation (1) with $\gamma_r = 1$. The objective is strictly concave in c_r , so the first order condition characterizes the optimum on the relevant domain $c_r \in [0, d_p]$; beyond d_p , additional severity produces no further deterrence, since $\hat{\tau}$ is already zero, and only adds public opinion costs. The first order condition is

$$\frac{B_s + \theta_0}{d_p} = 2\theta_1 c_r,$$

which reads: the marginal benefit of severity, deterrence at rate $1/d_p$ valued at the status quo prize plus the production cost saved per deterred protest, equals the marginal public opinion

³Their government minimizes $\theta^j(c) + C(c)$; the uniform planning prior makes the probability of regime change equal to the threshold itself, which is the same simplification used here.

cost. Solving,

$$c_r^* = \min \left\{ \frac{B_s + \theta_0}{2\theta_1 d_p}, d_p \right\}. \quad (2)$$

Proposition 2 (The optimal severity policy). *The state's objective is strictly concave on $[0, d_p]$ and severity above d_p is never optimal, so the optimum is unique and given by (2). The solution is interior if and only if $B_s + \theta_0 < 2\theta_1 d_p^2$; otherwise the state sits at the full-deterrence corner $c_r^* = d_p$, at which the protest never succeeds. In either case $c_r^* > 0$.*

The full proof is in the appendix. Panel A of Figure 1 maps the two regimes in the parameter space. At the interior optimum, the equilibrium success threshold, which is also the probability of policy change and the expected protest size, is

$$\hat{\tau}^* = 1 - \frac{B_s + \theta_0}{2\theta_1 d_p^2}.$$

3.5 Comparative statics.

The comparative statics at the interior solution are, without exception, unambiguous.

Proposition 3 (Comparative statics). *At the interior solution,*

$$\begin{aligned} \frac{\partial c_r^*}{\partial B_s} &= \frac{1}{2\theta_1 d_p} > 0, & \frac{\partial c_r^*}{\partial \theta_0} &= \frac{1}{2\theta_1 d_p} > 0, & \frac{\partial c_r^*}{\partial \theta_1} &= -\frac{B_s + \theta_0}{2\theta_1^2 d_p} < 0, \\ \frac{\partial c_r^*}{\partial d_p} &= -\frac{B_s + \theta_0}{2\theta_1 d_p^2} < 0, & \frac{\partial c_r^*}{\partial B_p} &= 0, \end{aligned}$$

and the equilibrium success probability $\hat{\tau}^$ is decreasing in B_s and θ_0 , increasing in θ_1 and d_p , and independent of B_p .*

Figure 1 illustrates these comparative statics by region of the parameter space, plotting the latent interior solution against the effective policy in each regime.

The signs admit direct interpretation. An increase in the production cost parameter θ_0 raises the chosen severity: if repressing each protester is dearer, the state deters harder to face fewer of them. An increase in the public opinion cost parameter θ_1 lowers severity. An increase in the intrinsic motivation d_p lowers severity, because motivated citizens are expensive to deter; note,

Region-conditional comparative statics, global-games version
 $(\theta_1 = 1, d_p = 3 \text{ except panel C; } B_s = 3 \text{ in panel C; dotted = latent interior FOC, solid = effective policy})$

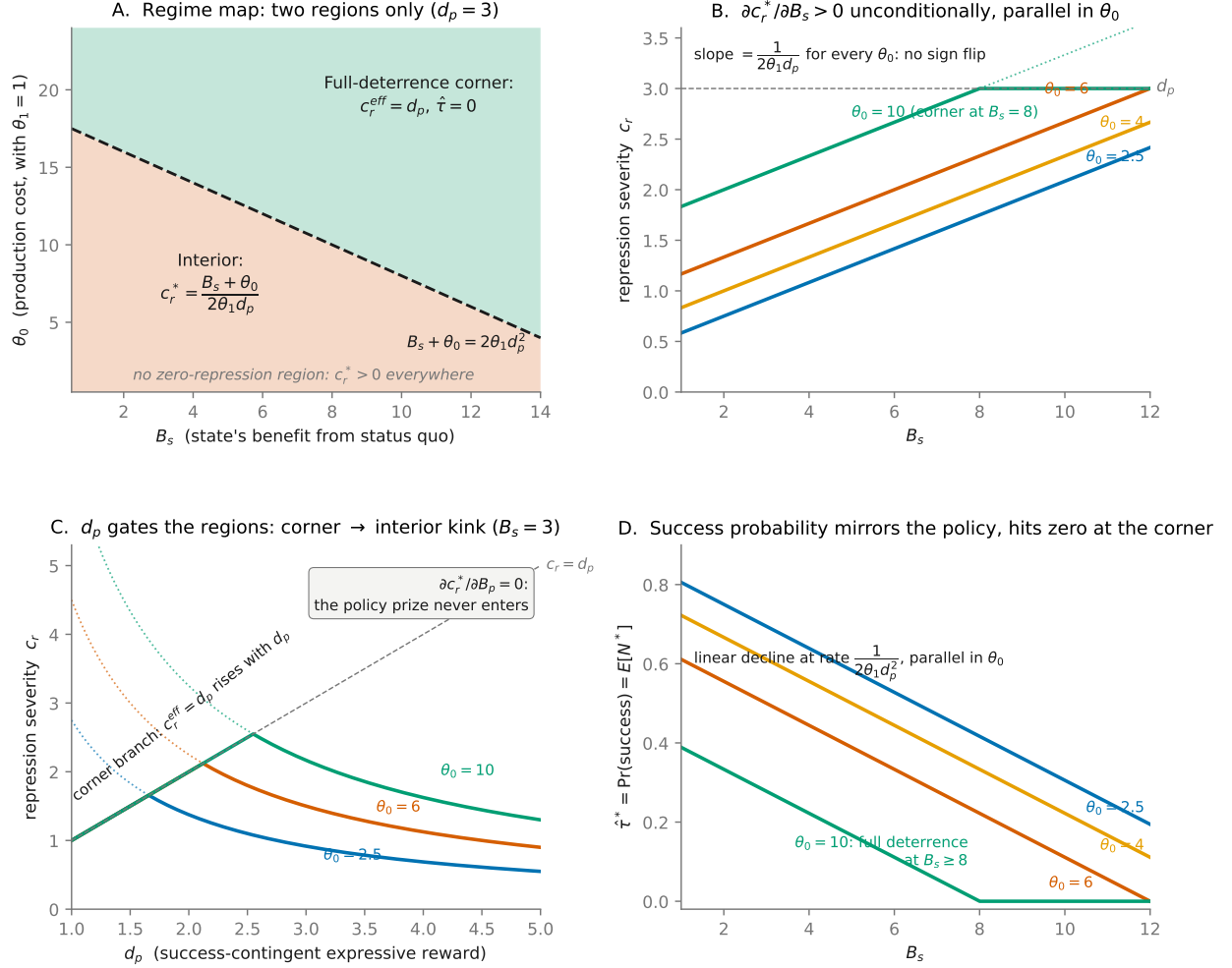


Figure 1: Region-conditional comparative statics of the repression severity policy ($\theta_1 = 1$; $d_p = 3$ except in panel C; $B_s = 3$ in panel C). Panel A maps the two regions of the parameter space: the interior solution and the full-deterrence corner, separated by $B_s + \theta_0 = 2\theta_1 d_p^2$; there is no zero-repression region. Panel B shows that c_r^* rises linearly in B_s with slope $1/(2\theta_1 d_p)$ for every θ_0 , flattening at the corner $c_r = d_p$. Panel C shows the kinked, non-monotone effect of d_p : the effective policy rides the corner $c_r = d_p$ upward while full deterrence is cheap, then switches to the declining interior solution. Panel D plots the equilibrium success probability $\hat{\tau}^*$, which declines linearly in B_s at rate $1/(2\theta_1 d_p^2)$ and reaches zero at the corner. Dotted curves are the latent interior first order condition; solid curves are the effective policy.

however, that here the state's retreat does not fully compensate, since $\hat{\tau}^*$ is increasing in d_p , so more motivated citizenries both face milder repression and succeed more often.

Two features of these results deserve emphasis.

First, the effect of the status quo prize B_s is unambiguously positive, with no parametric condition attached. In coordination models resolved by equilibrium selection, comparative statics of this kind are typically regime-conditional, holding in some regions of the parameter space and reversing in others (in De Jaegher and Hoyer 2019, for instance, repression deters when $V > 2c$ and backfires when $V < 2c$). Here the unique threshold equilibrium makes the deterrence technology the linear relation (1), and the marginal value of severity is monotone in the stakes. Panels B and D of Figure 1 display this monotone response and its mirror image in the success probability.

Second, B_p never enters the state's problem. The severity policy does not reckon with the policy prize at all: positive severity is optimal whenever $B_s + \theta_0 > 0$, that is, always, and the size of the prize the citizens would win is irrelevant to how hard the state represses. Panel C of Figure 1 shows the parameter that gates the policy instead: the effective severity is non-monotone in d_p , riding the full-deterrence corner upward before switching to the declining interior solution. This is the model working as intended, and it delimits its scope. The model instantiates a mass setting in which individual participation is not pivotal; in a small-numbers setting, the pivotal calculus of a small group carries B_p into the participation margin and, with it, into the severity policy. Which margin governs actual protest participation is an empirical question, and the sharp difference in the role of B_p across the two margins converts that question into a testable one.

3.6 An extension: Repression, protests, and emotions.

It is possible to adapt the model and introduce the role of emotions in the game's dynamics. Let the expressive reward be a function of the emotional state " e " of an individual, $d_p(e) = d_0 + e(c_r)$. To the extent that, in democracies, open media allows citizens to learn about how repressive is the average action of a state operative, citizens may get angered when they learn about a high degree of severity of repression, so that $e'(c_r) > 0$ for low levels of c_r ; yet for high or extreme levels of c_r , anticipatory fear may dominate and $e'(c_r) < 0$.

In the global games framework this channel acquires a precise transmission mechanism: the

equilibrium threshold becomes

$$\hat{t}(c_r) = 1 - \frac{\gamma_r c_r}{d_0 + e(c_r)},$$

so the emotional response enters the citizens' stage through the denominator of the normalized participation cost. In what follows, I develop the extension in three steps. First, I derive the general condition under which severity deters or backfires. Next, I solve the state's problem when anger responds linearly to severity. Then, I consider an outrage specification in which the anger response accelerates, which produces a backfire region and caps the state's optimal severity.

Let us fix $\gamma_r = 1$ and differentiate the threshold with respect to the severity policy. We get

$$\frac{\partial \hat{t}}{\partial c_r} = - \frac{d_0 + e(c_r) - c_r e'(c_r)}{(d_0 + e(c_r))^2}.$$

Define the elasticity of motivation with respect to severity as $\varepsilon(c_r) = c_r e'(c_r)/d_p(c_r)$.

Proposition 4 (Deterrence and backlash). *Severity deters, $\partial \hat{t}/\partial c_r < 0$, if and only if $\varepsilon(c_r) < 1$; it backfires, $\partial \hat{t}/\partial c_r > 0$, if and only if $\varepsilon(c_r) > 1$.*

In other words, whether repression discourages or recruits is not a psychological primitive but an elasticity condition on the anger response. Note that this gives the “punishment puzzle” (Davenport 2007a) a precise formulation: the deterrence and backlash findings that coexist in the empirical literature (della Porta 1995; Christopher Michael Sullivan and Davenport 2017; Adam-Troian, Çelebi, and Mahfud 2020) correspond to the two sides of the same inequality.

Let us begin with the linear case, $e(c_r) = \alpha c_r$ with $\alpha \geq 0$, so that $d_p(c_r) = d_0 + \alpha c_r$. Note that $\varepsilon(c_r) = \alpha c_r/(d_0 + \alpha c_r) < 1$ for every severity level, so linear anger never reverses deterrence; it dampens it, given that $\partial \hat{t}/\partial c_r = -d_0/(d_0 + \alpha c_r)^2$. More importantly, the threshold is bounded below: as c_r grows, $\hat{t}$ approaches $1 - 1/\alpha$. Thus, if $\alpha \geq 1$, full deterrence is unattainable at any finite severity, and an angry-enough citizenry retains a strictly positive probability of policy change no matter how harsh the policy. If $\alpha < 1$, full deterrence requires $c_r = d_0/(1 - \alpha)$, which grows without bound as α approaches one.

The state's problem remains strictly concave, and the first order condition becomes $(B_s + \theta_0)d_0 = 2\theta_1 c_r(d_0 + \alpha c_r)^2$, which has a unique positive solution.⁴ The full-deterrence corner is

⁴The left-hand side is a positive constant and the right-hand side is strictly increasing from zero, so the two cross exactly once. The derivations for this section are collected in the appendix.

optimal if and only if $(B_s + \theta_0)(1 - \alpha)^3 \geq 2\theta_1 d_0^2$, which collapses to the condition of the baseline model at $\alpha = 0$ and becomes impossible to satisfy as α approaches one. Two comparative statics follow. First, $\partial c_r^*/\partial \alpha < 0$: facing a more anger-reactive citizenry, the state tempers the severity of repression. Second, $d\hat{\tau}^*/d\alpha > 0$: the equilibrium probability of policy change increases in anger-reactivity through both channels, directly, because anger blunts deterrence, and strategically, because the state represses less.

Proposition 5 (Linear anger). *Let $e(c_r) = \alpha c_r$ with $\alpha \geq 0$. Then: (i) $\varepsilon(c_r) < 1$ at every severity and $\partial \hat{\tau}/\partial c_r = -d_0/(d_0 + \alpha c_r)^2 < 0$; (ii) $\hat{\tau}(c_r) > 1 - 1/\alpha$ at every finite severity, so full deterrence is unattainable whenever $\alpha \geq 1$, while for $\alpha < 1$ it requires $c_r = d_0/(1 - \alpha)$; (iii) the state's objective is strictly concave, the optimum is unique, and the full-deterrence corner is optimal if and only if $(B_s + \theta_0)(1 - \alpha)^3 \geq 2\theta_1 d_0^2$; (iv) at the interior optimum, $\partial c_r^*/\partial \alpha < 0$ and $d\hat{\tau}^*/d\alpha > 0$.*

Next, let us consider an outrage specification, $e(c_r) = \alpha c_r^k$ with $k > 1$, in which the anger response accelerates with severity. The idea is that extreme severity is qualitatively different from routine crowd control: a police killing that saturates media coverage recruits participants in a way that tear gas does not (Reny and Newman 2021). Now $\varepsilon(c_r)$ is increasing in c_r and crosses one at $\tilde{c} = (d_0/(\alpha(k - 1)))^{1/k}$, so the threshold $\hat{\tau}$ is U-shaped: severity deters below $\tilde{c}$ and backfires above it. Two consequences follow. First, the backfire point caps the state's policy, and the optimal severity never exceeds $\tilde{c}$, given that pushing beyond it raises both the probability of policy change and the costs of repression. Second, if $\hat{\tau}(\tilde{c}) = 1 - \tilde{c}(k - 1)/(d_0 k) > 0$, full deterrence is unattainable at any severity, and the best the state can do is push the probability of policy change down to its floor at the backfire point.

Proposition 6 (Outrage). *Let $e(c_r) = \alpha c_r^k$ with $\alpha > 0$ and $k > 1$. Then: (i) $\hat{\tau}$ is U-shaped in c_r , with its minimum at $\tilde{c} = (d_0/(\alpha(k - 1)))^{1/k}$, where $\hat{\tau}(\tilde{c}) = 1 - \tilde{c}(k - 1)/(d_0 k)$; (ii) the optimal severity satisfies $c_r^* \leq \tilde{c}$; (iii) if $\hat{\tau}(\tilde{c}) > 0$, no severity attains full deterrence.*

Finally, fear is the mirror case and requires little development. If $e'(c_r) < 0$, then $\varepsilon(c_r) < 0$, deterrence is amplified, and the qualitative structure of the baseline model is preserved with a steeper deterrence derivative. The interesting asymmetry is that anger changes the structure of the state's problem, bounding its policy and possibly its payoff, while fear only rescales it. A full

nonmonotone specification, in which anger dominates at low severity and fear at extreme severity, combines the two regimes; its analysis adds no closed forms and is left for future work.

4 Empirical implications.

The game-theoretic model developed in the latter section offers several empirical implications. In this section, I suggest some of them and propose ways in which we would try to estimate the empirical relation between protests and repression.

First, the model suggests that in any empirical estimation of either the effect of protests on repression or vice versa, a naive regression between the two will give wrong estimates, even if it has many meaningful controls to account for omitted variables. This happens because repression and the size of protests are simultaneously defined. Hence, empirical research should consider using structural estimation models or models that exploit some natural experiment, such as instrumental variables. The incomplete-information structure strengthens the case for the structural route: because the equilibrium is unique, the model delivers point predictions, a likelihood can be written down for observed outcomes, and the parameters $(d_p, \theta_0, \theta_1, B_s)$ are identified from variation in repression severity and protest success without an auxiliary equilibrium selection assumption. Multiplicity is precisely what makes structural estimation of coordination models treacherous, and it is what the machinery removes.

Second, one contribution of the model is to bring attention that, in democracies, the production of repression is costly. Even if we are unsure about the magnitude of public opinion costs, the level of overall repression has production costs such as spending on police trucks, ammunition, armored water cannon trucks, police office buildings, and others. To the extent that fiscal spending has opportunity costs, given that the incumbent politicians could have utilized such resources directly favoring voters, the degree of severity of repression will have nontrivial strategic considerations.

On the other hand, although it is not a particularly surprising result, an increase in public opinion marginal costs may reduce the severity of repression. Hence, if such an increase is in place, as it is common in democracies, this could induce a bigger size of protests or more protests participation in general.

Moreover, suppose empirical researchers can track exogenous shocks that may influence either the production costs of the level of repression (in the model, θ_0) or shocks that influence the public opinion costs (in the model, θ_1), that are unrelated with the determinants of protest participation. Then, it should be possible to operationalize a measure of these shocks and exploit it under an instrumental variables model.

The model's distinctive predictions supply an additional, sharper test. The severity of repression should respond positively to the incumbent's stakes in the status quo (B_s) everywhere in the parameter space, and it should not respond to the size of the prize citizens would obtain from the policy change (B_p). The second prediction is the signature of the mass-mobilization margin: it holds because no participant is pivotal, and a pivotal calculus would restore B_p to the participation margin.

Existing cross-national evidence is consistent with the first of these. Klein and Regan (2018) define the concession cost of a protest as the political cost of altering the status quo policy in response to protesters' demands, which is the empirical counterpart of B_s ; across 161 countries between 1990 and 2014 they find that governments facing high concession costs respond coercively. Their dependent variable is the mode of the response, coercion or accommodation, and not the severity of repression on a continuous scale, so the correspondence is conceptual. The sign, however, is the one the model predicts everywhere in the parameter space. Settings where protests are massive and individual pivotality is negligible should conform to this pattern; settings with small, organized, pivotal groups should not. A design comparing repression severity across policy disputes that differ in the dispersion of the prospective benefits, holding the incumbent's stakes fixed, would discriminate between the two margins.

The informational structure also generates implications of its own. Citizens' behavior depends on their beliefs about the critical threshold, and those beliefs respond to public information about expected turnout, past protest sizes, and logistical focal points. This connects directly to Little's (2016) distinction between political and tactical coordination: information that improves citizens' assessments of the crowd required, and of the crowd forthcoming, operates on the very fundamental τ of the present model, and his result that better tactical information systematically increases protest suggests that communication technology should compress the noise σ and sharpen the participation cascade around the threshold. Empirically, exogenous variation in connectivity or

in media coverage of expected turnout could be used to test whether protest outcomes become more bimodal, either massive or absent, as the model predicts they should when strategic uncertainty shrinks.

However, one obvious and nontrivial challenge is how to measure any of these variables. First, although not easy to measure, protest size seems like the less difficult one of them all. Several data sets code protest event characteristics by analyzing media content. In some cases, protest size participation is recorded.⁵ Additionally, using survey data, one can approximate the intrinsic motivations to participate in protests (d_p) at the individual level by using question items that gauge political involvement or attitudes towards political participation. Note that under the success-contingent interpretation adopted here, the relevant survey construct is not generic expressiveness but the anticipated satisfaction of having contributed to a victory, which is closer to efficacy and pleasure-in-agency item batteries. The variable B_p , a fundamentally unobserved construct and among the hardest in the model to measure, exits the participation margin here, which is a modest empirical convenience of this modeling strategy.

On the other hand, it may be possible to proxy for the variables that measure the state's marginal costs of repression. For example, one could measure the marginal costs of producing a level of repression (θ_0), measuring the average wage of the repressive operatives, or the percent of the GDP that the state devotes to repressive bureaus. Furthermore, we might use the distance of protest events to police stations (or the relevant repression bureau physical offices) as an instrument to the level of repression in a given protest event or episode. The idea is that if this distance is far, there is more cost in mobilizing the repressive operatives. Hence there is an incentive to increase the severity of repression. Following a repression severity policy defined at a high level, we would predict a smaller protest size.

On the other hand, the marginal costs related to public opinion may be proxied by the conditions of the competitive political environment. Yet, thinking about this variable should hint that we also must consider the institutional framework of any specific national or subnational political level. For example, if repressive operatives are managed at the subnational or local level, this could create different effects for incumbents at the national level. All in all, the idea is that if the

⁵For example, the Mass Mobilization Protest Data project, the Armed Conflict Location & Event Data Project, the Social Conflict Analysis Database, etc.

relevant electoral setting is competitive, repression may be more costly for the incumbent.

Next, I briefly discuss ways in which we could measure the repression severity policy in democracies. In this case, data is indeed scarce. Although efforts are growing, such as the Mapping Police Violence project in the U.S., there is still much to be done. Yet, we still may want to think about how we could approach this construct if we had the data. At least, we could think of two ways to measure this. First, we could calculate the rate of victims of repression over the population as some studies have done. The key open question in this approach would be how to choose the approach geographical area to define the base of the rate. Second, if we had such data, we could elaborate on a measure of physical or psychological trauma for the average victim of repression.

Finally, using survey data or experiments, it is possible to measure the relation between the severity of repression and the emotional state it elicits in citizens that may or may not plan to participate in future protests. One important aspect of such empirical assessment would be to test how different levels of repression severity may have nonlinear effects on the emotional state of individuals. For example, one may probe subjects with different and explicit consequences of repression victimization under a survey experiment. Then, one would measure their emotional state. Finally, one would ask individuals' dispositions or attitudes to participate in protests. Under the lens of the extension, the object to estimate is the elasticity of intrinsic motivation with respect to severity: the model predicts deterrence where it is below one and backlash where it is above one, so the design just described doubles as a direct measurement of the backfire point.

5 Conclusion.

Using a game-theoretic model, this paper explored the relationship between repression and protests motivated by policy change in democratic regimes. To accomplish this, a simple sequential game is elaborated that considers citizens' decisions to protest and the state's decision to set a policy of severity of repression to deter protest participation. The citizens' stage is developed as a global game of policy change: the critical threshold that protests must reach is an uncertain fundamental, citizens hold noisy private assessments of it, and the resulting incomplete information selects a unique equilibrium in threshold strategies where the complete-information benchmark exhibits multiple equilibria.

In this setting, the state moves first and defines a degree of severity of repression policy, which sets the individual costs for the repressed protester. The state would prefer that protests be minimal to keep the policy status quo unchanged, yet producing repression is costly for the state. In the second stage, citizens decide whether to protest under strategic uncertainty about the critical threshold. The protest succeeds, and the incumbent politicians are forced to change the policy, exactly when the realized threshold falls below an equilibrium cutoff that the state's severity policy shifts down at a rate inversely proportional to citizens' intrinsic motivation.

The model offers several insights. First, the severity of repression and the prospects of protest are interlinked through a unique equilibrium, so the model's comparative statics are unconditional: severity rises with the incumbent's stakes in the status quo and with the marginal production cost of repression, and falls with the public opinion cost of severity and with citizens' intrinsic motivation. Second, the policy prize drops out of the participation margin altogether: in a mass protest no citizen is pivotal, so the prize cannot move the marginal participant. The contrast between this prediction and the pivotal logic of small-numbers settings is a testable difference between mass and small-numbers mobilization rather than a defect of the model. Third, increasing the marginal costs of the production level of repression increases the state's chosen degree of repression severity: deterrence economizes on the production of repression. On the other hand, increasing the marginal costs of negative public opinion due to repression makes the state reduce the severity of the repression policy.

A number of key open questions remain unexplored and suggest extensions to the model. First, the emotions extension develops the anger channel for linear and accelerating responses, showing that backlash is an elasticity condition and that outrage caps the state's optimal severity; a full nonmonotone specification, in which fear overtakes anger at extreme severity, remains open. Second, a repeated version in which citizens observe protest outcomes and update about a persistent threshold would connect the model to learning dynamics over protest waves. Third, the state itself could be given instruments over the information structure, censoring or publicizing turnout information, which the global games literature on communication technology suggests is a first-order policy margin (Little 2016). Overall, it seems that the biggest empirical challenge rests on how to measure key variables of the model. For example, one could measure the marginal production costs of the level of repression by measuring the average wage of repressive operatives

or the percentage of GDP that the state allocates to repressive bureaus. On the other hand, the degree of severity of repression could be measured by the physical or psychological trauma of the average repressed protester. Nevertheless, it is likely that a more feasible option is just measuring the number of repressed protesters as a rate of the total local population. In any case, researchers should continue making efforts to understand repression and protests since these have become highly salient and influential phenomena in current politics, even in developed democracies.

References

- [1] Adam-Troian, Jais, Elif Çelebi, and Yara Mahfud. 2020. “‘Return of the Repressed’: Exposure to Police Violence Increases Protest and Self-Sacrifice Intentions for the Yellow Vests.” *Group Processes & Intergroup Relations* 23(8): 1171–86.
- [2] Aytaç, S. Erdem, Luis Schiumerini, and Susan Stokes. 2017. “Protests and Repression in New Democracies.” *Perspectives on Politics* 15(1): 62–82.
<https://doi.org/10.1017/S1537592716004138>.
- [3] Belur, Jyoti. 2010. “Police and the Use of Deadly Force.” In *Permission to Shoot? Police Use of Deadly Force in Democracies*, ed. Jyoti Belur. New York, NY: Springer New York, 1–20.
https://doi.org/10.1007/978-1-4419-0975-6_1.
- [4] Bor, Jacob, Atheendar S Venkataramani, David R Williams, and Alexander C Tsai. 2018. “Police Killings and Their Spillover Effects on the Mental Health of Black Americans: A Population-Based, Quasi-Experimental Study.” *The Lancet* 392(10144): 302–10.
- [5] Bosi, Lorenzo, Marco Giugni, and Katrin Uba. 2016. “The Consequences of Social Movements: Taking Stock and Looking Forward.” In *The Consequences of Social Movements*, eds. Lorenzo Bosi, Marco Giugni, and Katrin Uba. Cambridge: Cambridge University Press, 3–38.
- [6] Bueno de Mesquita, Ethan. 2010. “Regime Change and Revolutionary Entrepreneurs.” *American Political Science Review* 104(3): 446–66.
- [7] Bueno de Mesquita, Ethan, and Mehdi Shadmehr. 2023. “Rebel Motivations and Repression.” *American Political Science Review* 117(2): 734–50.

- [8] Carey, Sabine C. 2006. "The Dynamic Relationship between Protest and Repression." *Political Research Quarterly* 59(1): 1–11.
- [9] Carlsson, Hans, and Eric van Damme. 1993. "Global Games and Equilibrium Selection." *Econometrica* 61(5): 989–1018.
- [10] Çelebi, Elif, Jais Adam-Troian, and Yara Mahfud. 2020. "Positive Links between Exposure to Police Violence, PTSD, and Depression Symptoms among Yellow Vests Protesters in France." *Journal of Interpersonal Violence*: 0886260520935863.
- [11] Chenoweth, Erica, Evan Perkoski, and Sooyeon Kang. 2017. "State Repression and Nonviolent Resistance." *Journal of Conflict Resolution* 61(9): 1950–69.
- [12] Cooper, Hannah, Lisa Moore, Sofia Gruskin, and Nancy Krieger. 2004. "Characterizing Perceived Police Violence: Implications for Public Health." *American Journal of Public Health* 94(7): 1109–18. <https://pubmed.ncbi.nlm.nih.gov/15226128>.
- [13] Curtice, Travis B., and Brandon Behlendorf. 2021. "Street-Level Repression: Protest, Policing, and Dissent in Uganda." *Journal of Conflict Resolution* 65(1): 166–94. <https://doi.org/10.1177/0022002720939304>.
- [14] Davenport, Christian. 2007a. "State Repression and Political Order." *Annual Review of Political Science* 10: 1–23. <https://doi.org/10.1146/annurev.polisci.10.101405.143216>.
- [15] ———. 2007b. *State Repression and the Domestic Democratic Peace*. Cambridge University Press.
- [16] Davenport, Christian, and David A. Armstrong. 2004. "Democracy and the Violation of Human Rights: A Statistical Analysis from 1976 to 1996." *American Journal of Political Science* 48(3): 538–54. <https://onlinelibrary.wiley.com/doi/full/10.1111/j.0092-5853.2004.00086.x> (November 9, 2021).
- [17] De Jaegher, Kris, and Britta Hoyer. 2019. "Preemptive Repression: Deterrence, Backfiring, Iron Fists, and Velvet Gloves." *Journal of Conflict Resolution* 63(2): 502–27. <https://doi.org/10.1177/0022002717750450>.

- [18] deMeritt, Jacqueline H R. 2016. "The Strategic Use of State Repression and Political Violence." Oxford Research Encyclopedia of Politics. Oxford University Press. <https://oxfordre.com/politics/view/10.1093/acrefore/9780190228637.001.0001/acrefore-9780190228637-e-32>.
- [19] DeVyllder, Jordan E et al. 2017. "Prevalence, Demographic Variation and Psychological Correlates of Exposure to Police Victimization in Four US Cities." *Epidemiology and Psychiatric Sciences* 26(5): 466–77.
- [20] DeVyllder, Jordan, Lisa Fedina, and Bruce Link. 2020. "Impact of Police Violence on Mental Health: A Theoretical Framework." *American Journal of Public Health* 110(11): 1704–10. <https://ajph.aphapublications.org/doi/abs/10.2105/AJPH.2020.305874> (November 14, 2021).
- [21] Ekins, Emily E. 2016. "Policing in America: Understanding Public Attitudes toward the Police. Results from a National Survey." *Results from a National Survey* (December 7, 2016).
- [22] Fingerhut, Hannah. 2017. *Deep Racial, Partisan Divisions in Americans' Views of Police Officers*. Pew Research Center.
- [23] Gehlbach, Scott, Konstantin Sonin, and Milan W. Svolik. 2016. "Formal Models of Nondemocratic Politics." *Annual Review of Political Science* 19: 565–84.
- [24] Ginkel, John, and Alastair Smith. 1999. "So You Say You Want a Revolution: A Game Theoretic Explanation of Revolution in Repressive Regimes." *The Journal of Conflict Resolution* 43(3): 291–316.
- [25] Hindelang, Michael J. 1974. "Public Opinion Regarding Crime, Criminal Justice, and Related Topics." *Journal of Research in Crime and Delinquency* 11(2): 101–16.
- [26] Klein, Graig R., and Patrick M. Regan. 2018. "Dynamics of Political Protests." *International Organization* 72(2): 485–521. <https://doi.org/10.1017/S0020818318000061>.
- [27] Little, Andrew T. 2016. "Communication Technology and Protest." *The Journal of Politics* 78(1): 152–66.

- [28] Machado, Fabiana, Carlos Scartascini, and Mariano Tommasi. 2011. "Political Institutions and Street Protests in Latin America." *Journal of Conflict Resolution* 55(3): 340–65. <https://doi.org/10.1177/0022002711400864>.
- [29] Morris, Stephen, and Hyun Song Shin. 1998. "Unique Equilibrium in a Model of Self-Fulfilling Currency Attacks." *American Economic Review* 88(3): 587–97.
- [30] Morris, Stephen, and Hyun Song Shin. 2003. "Global Games: Theory and Applications." In *Advances in Economics and Econometrics: Theory and Applications, Eighth World Congress*, eds. Mathias Dewatripont, Lars Peter Hansen, and Stephen J. Turnovsky. Cambridge: Cambridge University Press, 56–114.
- [31] Nam, Taehyun. 2007. "Rough Days in Democracies: Comparing Protests in Democracies." *European Journal of Political Research* 46(1): 97–120. <https://doi.org/10.1111/j.1475-6765.2006.00645.x>.
- [32] Nordås, Ragnhild, and Christian Davenport. 2013. "Fight the Youth: Youth Bulges and State Repression." *American Journal of Political Science* 57(4): 926–40.
- [33] Pierskalla, Jan Henryk. 2010. "Protest, Deterrence, and Escalation: The Strategic Calculus of Government Repression." *Journal of Conflict Resolution* 54(1): 117–45.
- [34] della Porta, Donatella. 1995. "Social Movements and the State: Thoughts on the Policing of Protest." EUI Working Paper RSC No. 95/13. Florence: European University Institute, Robert Schuman Centre.
- [35] Reny, Tyler T, and Benjamin J Newman. 2021. "The Opinion-Mobilizing Effect of Social Protest against Police Violence: Evidence from the 2020 George Floyd Protests." *American Political Science Review* 115(4): 1499–1507. <https://www.cambridge.org/core/article/opinionmobilizing-effect-of-social-protest-against-police-violence-evidence-from-the-2020-george-floyd-protests/C62FCC5556A43F0AE3CDA5EB6AFD3673>.
- [36] Ritter, Emily Hencken. 2014. "Policy Disputes, Political Survival, and the Onset and Severity of State Repression." *Journal of Conflict Resolution* 58(1): 143–68.

- [37] Su, Yen-Pin. 2015. “Anti-Government Protests in Democracies: A Test of Institutional Explanations.” *Comparative Politics* 47(2): 149–67.
- [38] Sullivan, Christopher M. 2016. “Undermining Resistance: Mobilization, Repression, and the Enforcement of Political Order.” *Journal of Conflict Resolution* 60(7): 1163–90.
- [39] Sullivan, Christopher Michael, and Christian Davenport. 2017. “The Rebel Alliance Strikes Back: Understanding the Politics of Backlash Mobilization.” *Mobilization* 22(1): 39–56.
- [40] Whitehead, Neil L, Jo Ellen Fair, and Leigh A Payne. 2010. *Violent Democracies in Latin America*. Duke University Press.
- [41] Wood, Elisabeth Jean. 2003. *Insurgent Collective Action and Civil War in El Salvador*. Cambridge: Cambridge University Press.
- [42] Worden, Robert E. 2015. *2 The Causes of Police Brutality: Theory and Evidence on Police Use of Force*. Routledge, New York, NY, ed.
- [43] Zamble, Edward, and Phyllis Annesley. 1987. “Some Determinants of Public Attitudes toward the Police.” *Journal of Police Science & Administration*.

6 Appendix.

6.1 A preliminary lemma.

Lemma 1 (Posterior beliefs under the improper prior). *Suppose τ has the improper uniform prior on $\mathbb{R}$ and $x_i = \tau + \sigma \varepsilon_i$ with $\varepsilon_i \sim f$. Then the posterior density of τ given x_i is $f((x_i - \tau)/\sigma)/\sigma$, and for every evaluation point t ,*

$$\Pr(\tau \leq t \mid x_i) = 1 - F\left(\frac{x_i - t}{\sigma}\right).$$

Proof. By Bayes’ rule with a flat prior, the posterior density of τ given x_i is proportional to the likelihood of x_i given τ , which is $f((x_i - \tau)/\sigma)/\sigma$. This function integrates to one over $\tau \in \mathbb{R}$: substituting $u = (x_i - \tau)/\sigma$, so that $d\tau = -\sigma du$,

$$\int_{-\infty}^{\infty} \frac{1}{\sigma} f\left(\frac{x_i - \tau}{\sigma}\right) d\tau = \int_{-\infty}^{\infty} f(u) du = 1,$$

so the proportionality constant is one and the likelihood is itself the posterior density. For the distribution function, integrate over $\tau \leq t$ with the same substitution; the bound $\tau = -\infty$ maps to $u = \infty$ and $\tau = t$ maps to $u = (x_i - t)/\sigma$, so

$$\Pr(\tau \leq t \mid x_i) = \int_{-\infty}^t \frac{1}{\sigma} f\left(\frac{x_i - \tau}{\sigma}\right) d\tau = \int_{(x_i - t)/\sigma}^{\infty} f(u) du = 1 - F\left(\frac{x_i - t}{\sigma}\right). \quad \square$$

6.2 Proof of Proposition 1.

Proof of Proposition 1. Fix a cutoff strategy profile with cutoff $\hat{x}$: citizen i participates if and only if $x_i < \hat{x}$.

Step 1: participation and the induced success threshold. At true threshold τ , the mass of participants is

$$N^*(\tau) = \Pr(x_i < \hat{x} \mid \tau) = \Pr(\tau + \sigma \varepsilon_i < \hat{x}) = \Pr\left(\varepsilon_i < \frac{\hat{x} - \tau}{\sigma}\right) = F\left(\frac{\hat{x} - \tau}{\sigma}\right),$$

which is continuous and strictly decreasing in τ , with $N^* \rightarrow 1$ as $\tau \rightarrow -\infty$ and $N^* \rightarrow 0$ as $\tau \rightarrow \infty$. The protest succeeds when $N^*(\tau) \geq \tau$. The left side of this inequality is strictly decreasing in τ and the right side is strictly increasing, so there is exactly one crossing $\hat{\tau}$ solving $F((\hat{x} - \hat{\tau})/\sigma) = \hat{\tau}$, and success occurs exactly on $\tau \leq \hat{\tau}$.

Step 2: the marginal citizen. A citizen with signal x_i gains $\Pr(\tau \leq \hat{\tau} \mid x_i) d_p - \gamma_r c_r$ from participating. By Lemma 1 evaluated at $t = \hat{\tau}$ and $x_i = \hat{x}$,

$$\Pr(\tau \leq \hat{\tau} \mid x_i = \hat{x}) = 1 - F\left(\frac{\hat{x} - \hat{\tau}}{\sigma}\right) = 1 - \hat{\tau},$$

where the second equality substitutes the crossing condition from Step 1. Indifference of the marginal citizen requires $(1 - \hat{\tau}) d_p - \gamma_r c_r = 0$; solving,

$$1 - \hat{\tau} = \frac{\gamma_r c_r}{d_p} \iff \hat{\tau} = 1 - \frac{\gamma_r c_r}{d_p}.$$

Step 3: the cutoff signal. From the crossing condition, $F((\hat{x} - \hat{\tau})/\sigma) = \hat{\tau}$; since F is strictly increasing with full support, $(\hat{x} - \hat{\tau})/\sigma = F^{-1}(\hat{\tau})$, hence $\hat{x} = \hat{\tau} + \sigma F^{-1}(\hat{\tau})$.

Step 4: uniqueness within monotone strategies. It remains to show that only one cutoff is consistent with equilibrium. Differentiate the crossing condition implicitly with respect to $\hat{x}$, writing

$z = (\hat{x} - \hat{\tau}(\hat{x}))/\sigma$:

$$\frac{f(z)}{\sigma} (1 - \hat{\tau}'(\hat{x})) = \hat{\tau}'(\hat{x}) \implies \hat{\tau}'(\hat{x}) = \frac{f(z)/\sigma}{1 + f(z)/\sigma} \in (0, 1).$$

The marginal citizen's gain at cutoff $\hat{x}$ is $(1 - \hat{\tau}(\hat{x}))d_p - \gamma_r c_r$, which is strictly decreasing in $\hat{x}$ because $\hat{\tau}'(\hat{x}) > 0$. Hence at most one cutoff makes the marginal citizen indifferent, and the pair $(\hat{x}, \hat{\tau})$ from Steps 2 and 3 is the unique monotone equilibrium. The extension of uniqueness beyond monotone strategies follows from iterated deletion of dominated strategies, as in Morris and Shin (2003); see also the discussion in Bueno de Mesquita and Shadmehr (2023).

Step 5: boundary policies. If $\gamma_r c_r \geq d_p$, the participation gain satisfies $\Pr(\cdot)d_p - \gamma_r c_r \leq d_p - \gamma_r c_r \leq 0$ for every belief, with strict inequality whenever the belief is interior, so no citizen participates and $N^* = 0 < \tau$ for every $\tau > 0$. If $c_r = 0$, the gain is $\Pr(\cdot)d_p \geq 0$, participation is weakly dominant, every citizen participates, and the protest succeeds for every $\tau \leq 1$, that is, $\hat{\tau} = 1$. $\square$

Remark 1. A natural question about the proof above is why the indifference condition is solved leaving $\hat{\tau}$ as is, instead of solving for the fixed point of the belief consistency condition. The answer is that both conditions are used, and the fixed point is indeed solved; it simply delivers the unknown that the analysis does not display. Note that the equilibrium is a system of two equations in two unknowns, $(\hat{x}, \hat{\tau})$. The belief consistency condition alone cannot be solved as a standalone fixed point, given that it is one equation in two unknowns. In other words, for every candidate cutoff $\hat{x}$ it defines an induced threshold $\hat{\tau}(\hat{x})$, and the indifference condition provides the second relation; the equilibrium is the intersection of the two.

The reason the algebra collapses is that both conditions depend on $(\hat{x}, \hat{\tau})$ only through the statistic $z = (\hat{x} - \hat{\tau})/\sigma$. Under the improper uniform prior, the posterior probability that the protest succeeds is $\Pr(\tau \leq t \mid x_i = \hat{x}) = 1 - F((\hat{x} - t)/\sigma)$ for any evaluation point t . Hence, evaluating at $t = \hat{\tau}$ and substituting $F(z) = \hat{\tau}$ is not circular; it is the ordinary elimination of one unknown using the other equation. The system becomes triangular. First, the two conditions jointly yield $(1 - \hat{\tau})d_p = \gamma_r c_r$, which pins down $\hat{\tau} = 1 - \gamma_r c_r / d_p$. Then, the belief consistency condition, used again in the other direction, recovers the cutoff signal $\hat{x} = \hat{\tau} + \sigma F^{-1}(\hat{\tau})$, which is the fixed point in question. The main text does not display it because the state's problem only requires $\hat{\tau}$, and in the small-noise limit $\hat{x} \rightarrow \hat{\tau}$, so the distinction vanishes where the state's stage operates.

Finally, note that this triangular structure is a consequence of the improper prior. Under a

proper prior, the posterior does not reduce to $1 - F(z)$, the two conditions must be solved as a genuine two-dimensional fixed point, and no closed form obtains.⁶

6.3 Proof of Proposition 2.

Proof of Proposition 2. Substituting $\hat{\tau}(c_r) = 1 - c_r/d_p$ into the objective and expanding term by term,

$$u_S(c_r) = (1 - \hat{\tau})B_s - \theta_0\hat{\tau} - \theta_1c_r^2 = \frac{c_r}{d_p}B_s - \theta_0\left(1 - \frac{c_r}{d_p}\right) - \theta_1c_r^2 = \frac{B_s + \theta_0}{d_p}c_r - \theta_0 - \theta_1c_r^2.$$

Differentiating twice, $u'_S(c_r) = (B_s + \theta_0)/d_p - 2\theta_1c_r$ and $u''_S(c_r) = -2\theta_1 < 0$: the objective is strictly concave, so the first order condition characterizes the unique maximizer on any interval. Setting $u'_S = 0$ gives $c_r^\circ = (B_s + \theta_0)/(2\theta_1d_p)$. If $c_r^\circ \leq d_p$, equivalently $B_s + \theta_0 \leq 2\theta_1d_p^2$, then $c_r^* = c_r^\circ$; otherwise $u'_S > 0$ on all of $[0, d_p]$ and the maximum over the interval is at d_p . For $c_r > d_p$ the threshold is fixed at $\hat{\tau} = 0$, so $u_S(c_r) = B_s - \theta_1c_r^2$, which is strictly decreasing; severity above d_p is therefore never optimal, and $c_r^* = \min\{c_r^\circ, d_p\}$. Finally, $u'_S(0) = (B_s + \theta_0)/d_p > 0$, so $c_r^* > 0$ with no further parametric condition. $\square$

6.4 Proof of Proposition 3.

Proof of Proposition 3. At the interior solution $c_r^* = (B_s + \theta_0)/(2\theta_1d_p)$, differentiate with respect to each parameter in turn:

$$\frac{\partial c_r^*}{\partial B_s} = \frac{\partial}{\partial B_s} \left[\frac{B_s + \theta_0}{2\theta_1d_p} \right] = \frac{1}{2\theta_1d_p} > 0, \quad \frac{\partial c_r^*}{\partial \theta_0} = \frac{1}{2\theta_1d_p} > 0,$$

since the parameter enters the numerator linearly with unit coefficient in both cases;

$$\frac{\partial c_r^*}{\partial \theta_1} = \frac{B_s + \theta_0}{2d_p} \cdot \frac{\partial}{\partial \theta_1} [\theta_1^{-1}] = \frac{B_s + \theta_0}{2d_p} \cdot (-\theta_1^{-2}) = -\frac{B_s + \theta_0}{2\theta_1^2d_p} < 0,$$

$$\frac{\partial c_r^*}{\partial d_p} = \frac{B_s + \theta_0}{2\theta_1} \cdot \frac{\partial}{\partial d_p} [d_p^{-1}] = \frac{B_s + \theta_0}{2\theta_1} \cdot (-d_p^{-2}) = -\frac{B_s + \theta_0}{2\theta_1d_p^2} < 0;$$

⁶This is precisely why the improper prior is the standard device in this literature (Morris and Shin 2003; Bueno de Mesquita and Shadmehr 2023). At the critical threshold, the marginal citizen's belief about her rank in the population is uniform, so her success probability equals one minus the critical mass.

and B_p does not appear in the expression, so $\partial c_r^*/\partial B_p = 0$. For the success probability, substitute the interior solution into the threshold:

$$\hat{\tau}^* = 1 - \frac{c_r^*}{d_p} = 1 - \frac{B_s + \theta_0}{2\theta_1 d_p^2},$$

whence

$$\frac{\partial \hat{\tau}^*}{\partial B_s} = \frac{\partial \hat{\tau}^*}{\partial \theta_0} = -\frac{1}{2\theta_1 d_p^2} < 0, \quad \frac{\partial \hat{\tau}^*}{\partial \theta_1} = \frac{B_s + \theta_0}{2\theta_1^2 d_p^2} > 0, \quad \frac{\partial \hat{\tau}^*}{\partial d_p} = \frac{B_s + \theta_0}{\theta_1 d_p^3} > 0,$$

where the last derivative uses $\partial(d_p^{-2})/\partial d_p = -2d_p^{-3}$, so that $-\frac{B_s + \theta_0}{2\theta_1} \cdot (-2d_p^{-3}) = \frac{B_s + \theta_0}{\theta_1 d_p^3}$. B_p is again absent, so $\partial \hat{\tau}^*/\partial B_p = 0$. $\square$

6.5 Proof of Proposition 4.

Proof of Proposition 4. With $d_p(c_r) = d_0 + e(c_r)$ and $\gamma_r = 1$, apply the quotient rule to $\hat{\tau}(c_r) = 1 - c_r/d_p(c_r)$:

$$\frac{\partial \hat{\tau}}{\partial c_r} = -\frac{d_p(c_r) \cdot 1 - c_r e'(c_r)}{d_p(c_r)^2} = -\frac{d_p(c_r) - c_r e'(c_r)}{d_p(c_r)^2}.$$

The denominator is positive, so the derivative has the sign of $-(d_p(c_r) - c_r e'(c_r))$. Dividing the parenthesis by $d_p(c_r) > 0$,

$$d_p(c_r) - c_r e'(c_r) \geq 0 \iff 1 - \frac{c_r e'(c_r)}{d_p(c_r)} = 1 - \varepsilon(c_r) \geq 0,$$

so $\partial \hat{\tau}/\partial c_r < 0$ exactly when $\varepsilon(c_r) < 1$ and $\partial \hat{\tau}/\partial c_r > 0$ exactly when $\varepsilon(c_r) > 1$. $\square$

6.6 Proof of Proposition 5.

Proof of Proposition 5. Part (i). With $e(c_r) = \alpha c_r$, $e'(c_r) = \alpha$ and $d_p(c_r) = d_0 + \alpha c_r$, so

$$\varepsilon(c_r) = \frac{\alpha c_r}{d_0 + \alpha c_r} < 1 \iff \alpha c_r < d_0 + \alpha c_r \iff 0 < d_0,$$

which holds by assumption; by Proposition 4, severity always deters. For the derivative, the quotient-rule numerator is $(d_0 + \alpha c_r) - c_r \alpha = d_0$, hence $\partial \hat{\tau}/\partial c_r = -d_0/(d_0 + \alpha c_r)^2$.

Part (ii). Dividing numerator and denominator by c_r ,

$$\hat{\tau}(c_r) = 1 - \frac{c_r}{d_0 + \alpha c_r} = 1 - \frac{1}{d_0/c_r + \alpha} \rightarrow 1 - \frac{1}{\alpha} \quad \text{as } c_r \rightarrow \infty,$$

and the convergence is monotone from above by part (i), so $\hat{\tau}(c_r) > 1 - 1/\alpha$ at every finite severity. Full deterrence requires $\hat{\tau} = 0$, that is, $c_r = d_0 + \alpha c_r$, that is, $c_r(1 - \alpha) = d_0$. For $\alpha \geq 1$ the left side is nonpositive for every $c_r > 0$ while the right side is positive, so no finite severity attains it; for $\alpha < 1$ the unique solution is $c_r = d_0/(1 - \alpha)$.

Part (iii). Using $1 - \hat{\tau} = c_r/(d_0 + \alpha c_r)$, the objective is $u_S(c_r) = (B_s + \theta_0)c_r/(d_0 + \alpha c_r) - \theta_0 - \theta_1 c_r^2$. By the quotient rule,

$$\frac{d}{dc_r} \left[\frac{c_r}{d_0 + \alpha c_r} \right] = \frac{(d_0 + \alpha c_r) \cdot 1 - c_r \cdot \alpha}{(d_0 + \alpha c_r)^2} = \frac{d_0}{(d_0 + \alpha c_r)^2},$$

so $u'_S(c_r) = (B_s + \theta_0)d_0/(d_0 + \alpha c_r)^2 - 2\theta_1 c_r$; differentiating again, using $d[(d_0 + \alpha c_r)^{-2}]/dc_r = -2\alpha(d_0 + \alpha c_r)^{-3}$,

$$u''_S(c_r) = -\frac{2\alpha(B_s + \theta_0)d_0}{(d_0 + \alpha c_r)^3} - 2\theta_1 < 0,$$

so the objective is strictly concave. Moreover, $u'_S(0) = (B_s + \theta_0)d_0/d_0^2 = (B_s + \theta_0)/d_0 > 0$ and u'_S is strictly decreasing and diverges to $-\infty$, so u'_S has exactly one positive root, characterized by rearranging $u'_S = 0$ into $(B_s + \theta_0)d_0 = 2\theta_1 c_r(d_0 + \alpha c_r)^2$. For $\alpha < 1$ the relevant domain is $[0, \bar{c}]$ with $\bar{c} = d_0/(1 - \alpha)$ the full-deterrence corner from part (ii); by concavity, the corner is optimal exactly when $u'_S(\bar{c}) \geq 0$. Substituting $\bar{c}$: first,

$$d_0 + \alpha \bar{c} = d_0 + \frac{\alpha d_0}{1 - \alpha} = d_0 \frac{(1 - \alpha) + \alpha}{1 - \alpha} = \frac{d_0}{1 - \alpha},$$

hence

$$u'_S(\bar{c}) = (B_s + \theta_0)d_0 \cdot \frac{(1 - \alpha)^2}{d_0^2} - 2\theta_1 \cdot \frac{d_0}{1 - \alpha} = \frac{(B_s + \theta_0)(1 - \alpha)^2}{d_0} - \frac{2\theta_1 d_0}{1 - \alpha}.$$

Multiplying through by $d_0(1 - \alpha) > 0$, the condition $u'_S(\bar{c}) \geq 0$ is equivalent to $(B_s + \theta_0)(1 - \alpha)^3 \geq 2\theta_1 d_0^2$, as stated; the interior case is its strict complement. For $\alpha \geq 1$ there is no corner and the interior root is always the optimum; consistently, $(1 - \alpha)^3 \leq 0$ and the corner condition can never hold.

Part (iv). Write the first order condition as $G(c_r, \alpha) = (B_s + \theta_0)d_0(d_0 + \alpha c_r)^{-2} - 2\theta_1 c_r = 0$. The partial derivatives are

$$\frac{\partial G}{\partial \alpha} = (B_s + \theta_0)d_0 \cdot (-2)(d_0 + \alpha c_r)^{-3} \cdot c_r = -\frac{2(B_s + \theta_0)d_0 c_r}{(d_0 + \alpha c_r)^3} < 0, \quad \frac{\partial G}{\partial c_r} = u_S''(c_r) < 0,$$

so by the implicit function theorem $\partial c_r^* / \partial \alpha = -G_\alpha / G_{c_r} < 0$: a leading minus applied to the ratio of two negatives. For the threshold, $\hat{\tau}(c_r, \alpha) = 1 - c_r(d_0 + \alpha c_r)^{-1}$ depends on α both directly and through c_r^* :

$$\left. \frac{\partial \hat{\tau}}{\partial \alpha} \right|_{c_r} = -c_r \cdot (-1)(d_0 + \alpha c_r)^{-2} \cdot c_r = \frac{c_r^2}{(d_0 + \alpha c_r)^2} > 0, \quad \frac{\partial \hat{\tau}}{\partial c_r} = -\frac{d_0}{(d_0 + \alpha c_r)^2} < 0,$$

hence

$$\frac{d\hat{\tau}^*}{d\alpha} = \frac{\partial \hat{\tau}}{\partial \alpha} + \frac{\partial \hat{\tau}}{\partial c_r} \cdot \frac{\partial c_r^*}{\partial \alpha} > 0,$$

the first term positive and the second the product of two negatives. □

6.7 Proof of Proposition 6.

Proof of Proposition 6. Part (i). With $e(c_r) = \alpha c_r^k$, $e'(c_r) = \alpha k c_r^{k-1}$, and the numerator of $-\partial \hat{\tau} / \partial c_r$ from the proof of Proposition 4 is

$$d_p(c_r) - c_r e'(c_r) = d_0 + \alpha c_r^k - c_r \cdot \alpha k c_r^{k-1} = d_0 - \alpha(k-1)c_r^k,$$

which is strictly decreasing in c_r (as $k > 1$), positive at $c_r = 0$, and negative for large c_r . Its unique zero solves $\alpha(k-1)\tilde{c}^k = d_0$, that is, $\tilde{c} = (d_0/(\alpha(k-1)))^{1/k}$. Hence $\hat{\tau}$ is strictly decreasing on $(0, \tilde{c})$ and strictly increasing on $(\tilde{c}, \infty)$: U-shaped with its minimum at $\tilde{c}$. For the value at the minimum, use $\alpha\tilde{c}^k = d_0/(k-1)$ to compute the denominator,

$$d_0 + \alpha\tilde{c}^k = d_0 + \frac{d_0}{k-1} = d_0 \frac{(k-1)+1}{k-1} = \frac{d_0 k}{k-1},$$

so that

$$\hat{\tau}(\tilde{c}) = 1 - \frac{\tilde{c}}{d_0 k / (k-1)} = 1 - \frac{\tilde{c}(k-1)}{d_0 k}.$$

Part (ii). Write the objective as $u_S(c_r) = (B_s + \theta_0)(1 - \hat{\tau}(c_r)) - \theta_0 - \theta_1 c_r^2$. On $(\tilde{c}, \infty)$, part (i) gives $\hat{\tau}$ strictly increasing, so $1 - \hat{\tau}$ is strictly decreasing and the first term falls, while $-\theta_1 c_r^2$ also falls; hence u_S is strictly decreasing beyond $\tilde{c}$ and no $c_r > \tilde{c}$ can be optimal: $c_r^* \leq \tilde{c}$.

Part (iii). Full deterrence requires $\hat{\tau}(c_r) \leq 0$ at some severity. By part (i) the minimum of $\hat{\tau}$ over all severities is $\hat{\tau}(\tilde{c})$; if $\hat{\tau}(\tilde{c}) > 0$, then $\hat{\tau}(c_r) > 0$ at every severity and no policy attains full deterrence. □

All of the claims in this appendix were additionally verified numerically over randomized parameter draws.